

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-ALL]

TEST # 03
DATE : 30-04-2020
Test Type : Minor
Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry and Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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BEWARE OF NEGATIVE MARKING

1. A ring of radius R is charged uniformly with a charge $+Q$. The electric field at a point on its axis at a distance r from any point on the ring will be :-

(1) $\frac{KQ}{(r^2 - R^2)}$ (2) $\frac{KQ}{r^2}$
(3) $\frac{KQ}{r^3}(r^2 - R^2)^{1/2}$ (4) $\frac{KQr}{R^3}$

2. A spherical solid ball of volume V is made of a material of density ρ_1 . It is falling through a liquid of density ρ_2 ($\rho_2 < \rho_1$). Assume that the liquid applies a viscous force on the ball that is proportional to the square of its speed v , i.e., $F_{\text{viscous}} = -kv^2$ ($k > 0$). The terminal speed of the ball is :-

(1) $\sqrt{\frac{Vg(\rho_1 - \rho_2)}{k}}$ (2) $\frac{Vg\rho_1}{k}$
(3) $\sqrt{\frac{Vg\rho_1}{k}}$ (4) $\frac{Vg(\rho_1 - \rho_2)}{k}$

3. Two particles x and y have equal charges and possessing equal kinetic energy enter in a uniform magnetic field and describe circular path of radius of curvature r_1 and r_2 respectively. The ratio of their masses is :-

(1) $\left(\frac{r_1}{r_2}\right)$ (2) $\left(\frac{r_1}{r_2}\right)^{1/2}$
(3) $\left(\frac{r_1}{r_2}\right)^2$ (4) $\left(\frac{r_2}{r_1}\right)$

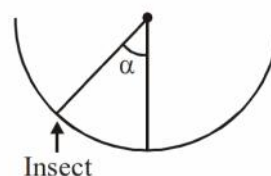
4. A gramophone record is revolving with an angular velocity ω . A coin is placed at a distance R from the centre of the record. The static coefficient of friction is μ . The coin will revolve with the record if :-

(1) $R > \frac{\mu g}{\omega^2}$ (2) $R = \frac{\mu g}{\omega^2}$ only
(3) $R < \frac{\mu g}{\omega^2}$ (4) $R \leq \frac{\mu g}{\omega^2}$

5. The phase difference corresponding to path difference of x is :

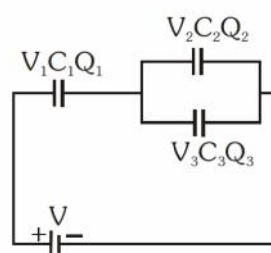
(1) $\frac{2\pi x}{\lambda}$ (2) $\frac{2\pi \lambda}{x}$
(3) $\frac{\pi x}{\lambda}$ (4) $\frac{\pi \lambda}{x}$

6. An insect crawls up a hemispherical surface very slowly (see the figure). The coefficient of friction between the insect and the surface is $1/3$. If the line joining the centre of the hemispherical surface to the insect makes an angle α with the vertical, the maximum possible value α is given by :-



(1) $\cot \alpha = 3$ (2) $\tan \alpha = 3$
(3) $\sec \alpha = 3$ (4) $\text{cosec } \alpha = 3$

7. In an adjoining figure three capacitors C_1 , C_2 and C_3 are joined to a battery. The correct condition will be :



(1) $Q_1 = Q_2 = Q_3$ and $V_1 = V_2 = V_3 = V$
(2) $Q_1 = Q_2 + Q_3$ and $V = V_1 + V_2 + V_3$
(3) $Q_1 = Q_2 + Q_3$ and $V = V_1 + V_2$
(4) $Q_2 = Q_3$ and $V_2 = V_3$

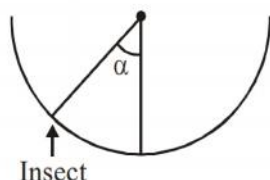
(Symbols have their usual meanings)

5. The phase difference corresponding to path difference of x is :

(1) $\frac{2\pi x}{\lambda}$ (2) $\frac{2\pi \lambda}{x}$

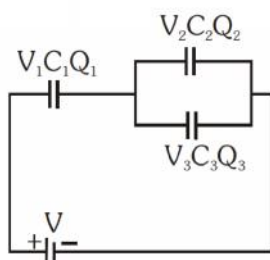
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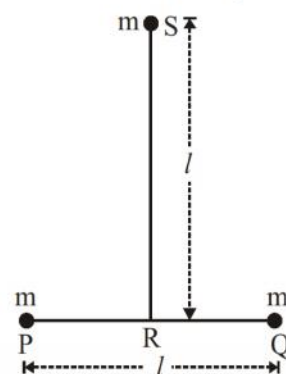
7. In an adjoining figure three capacitors C_1 , C_2 and C_3 are joined to a battery. The correct condition will be :



- (1) $Q_1 = Q_2 = Q_3$ and $V_1 = V_2 = V_3 = V$
(2) $Q_1 = Q_2 + Q_3$ and $V = V_1 + V_2 + V_3$
(3) $Q_1 = Q_2 + Q_3$ and $V = V_1 + V_2$
(4) $Q_2 = Q_3$ and $V_2 = V_3$

(Symbols have their usual meanings)

12. Two particles each of mass m are placed at points P and Q as shown in the figure. R is the mid-point of $PQ = l$. The gravitational force on the third particle of mass m placed at point S on the perpendicular bisector of PQ is :-

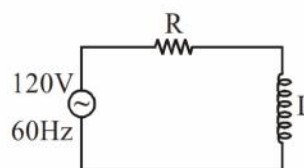


- (1) $\frac{Gm^2}{l^2}$ (2) $\frac{16Gm^2}{5l^2}$
(3) $\frac{16Gm^2}{5\sqrt{5}l^2}$ (4) $\frac{4\sqrt{2}Gm^2}{5l^2}$

13. A sphere of diameter r is cut from a sphere of radius r such that the centre of mass of the remaining mass be at maximum distance from original centre; then the distance is :-

- (1) $\frac{r}{2}$ (2) $\frac{r}{3}$
(3) $\frac{r}{6}$ (4) none of these

14. In the given circuit the potential difference across resistance is 54 V and power consumed by it is 16 W. If AC frequency is 60Hz find the value of L

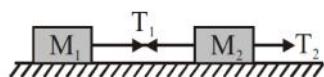


- (1) 1 H (2) 2 H (3) 5 H (4) 4 H

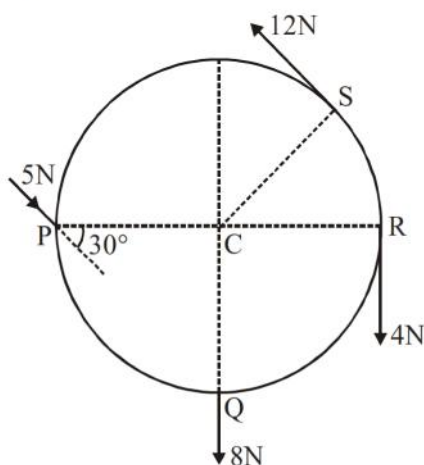
15. If the kinetic energy of a moving particle is E, then the De Broglie wavelength is :

- (1) $h\sqrt{2mE}$ (2) $\sqrt{\frac{2mE}{h}}$
(3) $\frac{h}{\sqrt{2mE}}$ (4) $\frac{hE}{\sqrt{2mE}}$

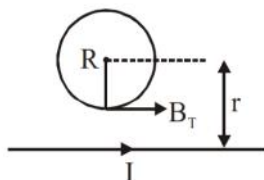
16. Two masses M_1 and M_2 are accelerated uniformly on a frictionless surface as shown in figure. The ratio of the tensions T_1/T_2 is :-



- (1) $\frac{M_1}{M_2}$ (2) $\frac{M_2}{M_1}$
(3) $\frac{(M_1 + M_2)}{M_2}$ (4) $\frac{M_1}{(M_1 + M_2)}$
17. Four forces are applied to a wheel of radius 10 cm as shown in the figure. The net torque produced by all these forces about C is :-



- (1) 1.25 Nm clockwise
(2) 1.25 Nm anticlockwise
(3) 1.05 Nm anticlockwise
(4) 1.05 Nm clockwise
18. In following diagram there is a straight wire carrying a current I . Consider a circular path with radius (R) near it. If $\vec{B}_T$ is the tangential component of magnetic field then find the value of integral $\int \vec{B}_T \cdot d\vec{l}$



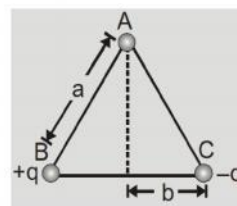
- (1) $\frac{\mu_0 I}{2\pi R}$ (2) $\frac{\mu_0 I}{4\pi R}$
(3) $\frac{\mu_0 I}{2\pi(r - R)}$ (4) Zero

19. The electric field part of an electromagnetic wave in a medium is represented by $E_x = 0$;

$$E_y = 2.5 \frac{N}{C} \cos \left[\left(2\pi \times 10^6 \frac{\text{rad}}{\text{m}} \right) t - \left(\pi \times 10^{-2} \frac{\text{rad}}{\text{s}} \right) x \right];$$

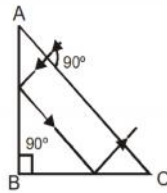
$E_z = 0$. The wave is :

- (1) Moving along $-x$ direction with frequency 10^6 Hz and wave length 200 m.
(2) Moving along y direction with frequency $2\pi \times 10^6$ Hz and wave length 200 m.
(3) Moving along x direction with frequency 10^6 Hz and wave length 100 m.
(4) Moving along x direction with frequency 10^6 Hz and wave length 200 m.
20. If $A = 100101$ and $B = 110110$ are applied to X-NOR gate, then the output will be :-
(1) 101100 (2) 011011
(3) 110111 (4) 010011
21. As shown in the fig. charges $+q$ and $-q$ are placed at the vertices B and C of an isosceles triangle. The potential at the vertex A is :-



- (1) $\frac{1}{4\pi\epsilon_0} \cdot \frac{2q}{\sqrt{a^2 + b^2}}$ (2) zero
(3) $\frac{1}{4\pi\epsilon_0} \cdot \frac{q}{\sqrt{a^2 + b^2}}$ (4) $\frac{1}{4\pi\epsilon_0} \cdot \frac{-q}{\sqrt{a^2 + b^2}}$
22. A rod of length l is hinged from one end. It is brought to a horizontal position and released. The angular velocity of the rod when it is in the vertical position is :-
(1) $\sqrt{\frac{g}{l}}$ (2) $\sqrt{\frac{2g}{l}}$
(3) $\sqrt{\frac{3g}{l}}$ (4) $\sqrt{\frac{g}{2l}}$
23. The minimum number of vectors of unequal magnitude required to produce a zero resultant is:-
(1) 2 (2) 3 (3) 4 (4) more than 4

24. A ray falls on a prism ABC (AB = BC) and travels as shown in figure. The minimum refractive index of the prism material should be

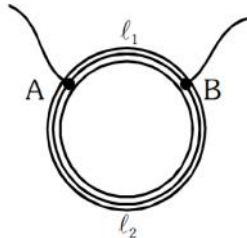


- (1) $\frac{4}{3}$ (2) $\sqrt{2}$
(3) 1.5 (4) $\sqrt{3}$
25. Atomic weight of a radioactive element is M_w gram. Radioactivity of m gram. of its mass is :-
(N_A = Avogadro number, λ = decay constant)

- (1) $N_A \lambda$ (2) $\left(\frac{N_A}{M_w} m\right) \lambda$
(3) $\left(\frac{N_A}{m}\right) \lambda$ (4) $\left(\frac{N_A M_w}{m}\right) \lambda$

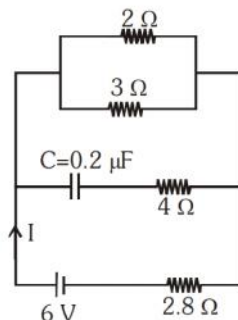
26. A ring is made of a wire having a resistance $R_0 = 12 \Omega$. Find the points A and B as shown in the figure at which a current carrying conductor should be connected so that the resistance R of the sub circuit between these points is equal to $\frac{8}{3} \Omega$:-

- (1) $\frac{\ell_1}{\ell_2} = \frac{3}{8}$
(2) $\frac{\ell_1}{\ell_2} = \frac{1}{2}$
(3) $\frac{\ell_1}{\ell_2} = \frac{5}{8}$
(4) $\frac{\ell_1}{\ell_2} = \frac{1}{3}$



27. In the given figure the steady state current I is :

- (1) zero
(2) 0.6 A
(3) 0.9 A
(4) 1.5 A



28. The relation between time t and distance x is
 $t = \alpha x^2 + \beta x$

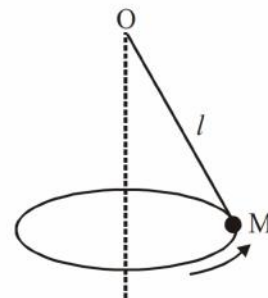
where α and β are constant. The retardation is :-

- (1) $2\alpha v^3$ (2) $2\beta v^3$ (3) $2\alpha\beta v^3$ (4) $2\beta^2 v^3$

29. A plano convex lens fits exactly into a plano concave lens. Their plane surfaces are parallel to each other. If lenses are made of different materials of refractive indices μ_1 and μ_2 and R is the radius of curvature of the curved surface of the lenses, then the focal length of combination is

- (1) $\frac{2R}{(\mu_2 - \mu_1)}$ (2) $\frac{R}{2(\mu_1 + \mu_2)}$
(3) $\frac{R}{2(\mu_1 - \mu_2)}$ (4) $\frac{R}{(\mu_1 - \mu_2)}$

30. A point mass m is suspended from a light thread of length l, fixed at O, is whirled in a horizontal circle at constant speed as shown. From your point of view, stationary with respect to the mass, the forces on the mass are :-



- (1)
- (2)
- (3)
- (4)

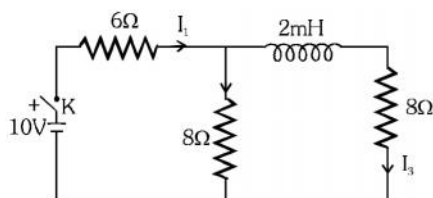
31. Two metal wires of identical dimensions are connected in series. If σ_1 and σ_2 are the conductivities of the metal wires respectively, the effective conductivity of the combination is :-

- (1) $\frac{\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$ (2) $\frac{2\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$ (3) $\frac{\sigma_1 + \sigma_2}{2\sigma_1 \sigma_2}$ (4) $\frac{\sigma_1 + \sigma_2}{\sigma_1 \sigma_2}$

32. If the earth were to suddenly contract to $1/n$ th of its present radius without any change in its mass, the duration (in hrs.) of the new day will be nearly:-

(1) $24/n$ (2) $24n$ (3) $24/n^2$ (4) $24n^2$

33. In the circuit shown in figure what is the value of I_1 just after pressing the key K ?



(1) $\frac{5}{7}$ A

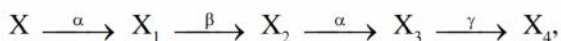
(2) $\frac{5}{11}$ A

(3) 1A (4) None of the above

34. A tuning fork vibrating with a sonometer having 20 cm wire produces 5 beats per sec. The beat frequency does not change if the length of the wire is changed to 21 cm. The frequency of the tuning fork must be :-

(1) 200 Hz (2) 210 Hz
(3) 205 Hz (4) 215 Hz

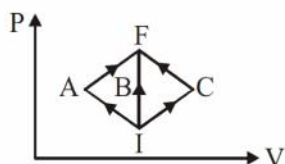
35. A radioactive nucleus decay as follows :-



if the atomic number and the mass number of X are 72 and 180 then the mass number and atomic number of X_4 are :-

(1) 172, 70 (2) 171, 69
(3) 172, 69 (4) 172, 68

36. In the P-V diagram, I is the initial state and F is the final state. The gas goes from I to F by (i) IAF, (ii) IBF, (iii) ICF. The heat absorbed by the gas is :-



(1) the same in all three processes
(2) the same in (i) and (ii)
(3) greater in (i) than in (ii)
(4) the same in (i) and (iii)

37. When a constant force is applied to a body moving with constant acceleration, power does not remain constant. For power to be constant, the force has to vary with speed as follows :-

(1) $F \propto \frac{1}{v}$

(2) $F \propto \frac{1}{\sqrt{v}}$

(3) $F \propto v$

(4) $F \propto v^2$

38. A particle is projected from the ground with an initial speed v at an angle θ with horizontal. The average velocity of the particle between its point of projection and highest point of trajectory is :-

(1) $\frac{v}{2} \sqrt{1+2\cos^2 \theta}$

(2) $\frac{v}{2} \sqrt{1+\cos^2 \theta}$

(3) $\frac{v}{2} \sqrt{1+3\cos^2 \theta}$

(4) $v \cos \theta$

39. A car sounding its horn at 480 Hz moves towards a high wall at a speed of 20 m/s. If the speed of sound is 340 m/s, the frequency of the reflected sound heard by the passenger sitting in the car will be the nearest to :-

(1) 480 Hz (2) 510 Hz
(3) 540 Hz (4) 570 Hz

40. Given that; $A = B = C$. If $\vec{A} + \vec{B} = \vec{C}$, then the angle between $\vec{A}$ and $\vec{C}$ is θ_1 . If $\vec{A} + \vec{B} + \vec{C} = 0$, then the angle between $\vec{A}$ and $\vec{C}$ is θ_2 . What is the relation between θ_1 and θ_2 ?

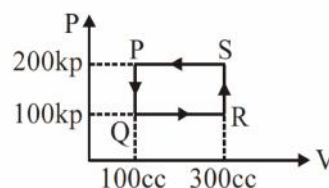
(1) $\theta_1 = \theta_2$

(2) $\theta_1 = \theta_2/2$

(3) $\theta_1 = 2\theta_2$

(4) None of these

41. A thermodynamic system is taken through the cycle PQRSP process. the net work done by the system is :-



(1) 20 J (2) -20 J (3) 400 J (4) -374 J

42. Force acting on a particle moving in a straight line varies with the velocity of the particle as $F = \frac{K}{v}$, where K is a constant. The work done by this force in time t is :-

- (1) $\frac{K}{v^2}t$ (2) $2Kt$
 (3) Kt (4) $\frac{2Kt}{v^2}$

43. An e.m.f. of 15 volt is applied in a circuit containing 5 henry inductance and 10 ohm resistance. The ratio of the currents at time $t \rightarrow \infty$ and at $t = 1$ second is:-

- (1) $\frac{e^{1/2}}{e^{1/2}-1}$ (2) $\frac{e^2}{e^2-1}$
 (3) $1-e^{-1}$ (4) e^{-1}

44. If intensity of each wave in the observed interference pattern in Young's double slit experiment is I_0 . then for some point P where the phase difference is ϕ , intensity I will be :

- (1) $I = I_0 \cos \phi$
 (2) $I = I_0 \cos^2 \phi$
 (3) $I = I_0 (1 + \cos \phi)$
 (4) $I = 2I_0 (1 + \cos \phi)$

45. A vector $\vec{Q}$ which has a magnitude of 8 is added to the vector $\vec{P}$ which lies along x-axis. The resultant of two vectors lies along y-axis and has magnitude twice that of $\vec{P}$. The magnitude of is $\vec{P}$:-

- (1) $\frac{6}{\sqrt{5}}$ (2) $\frac{8}{\sqrt{5}}$
 (3) $\frac{12}{\sqrt{5}}$ (4) $\frac{16}{\sqrt{5}}$

46. In d-block 3d series, which of the following element has maximum size ?

(1) Mn (2) Fe (3) Zn (4) Sc

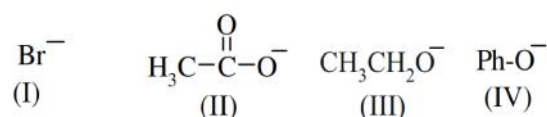
47. Geometrical and optical isomerism exhibited by :

(1) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ (2) $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$
 (3) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ (4) $[\text{Cr}(\text{OX})_3]^{3-}$

48. The major product obtained on interaction of phenol with sodium hydroxide and carbon dioxide is

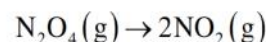
(1) salicylic acid (2) phthalic acid
 (3) benzoic acid (4) salicylaldehyde

49. Arrange the following in order of their leaving group tendency?



(1) $\text{I} > \text{II} > \text{III} > \text{IV}$ (2) $\text{I} > \text{III} > \text{II} > \text{IV}$
 (3) $\text{I} > \text{II} > \text{IV} > \text{III}$ (4) $\text{I} > \text{IV} > \text{III} > \text{II}$

50. What is the activation energy for the reverse of this reaction?



Data for the given reaction is

$\Delta H = +54 \text{ kJ}$ and $E_a = +57.2 \text{ kJ}$

(1) -54 kJ (2) $+3.2 \text{ kJ}$
 (3) $+60.2 \text{ kJ}$ (4) $+111.2 \text{ kJ}$

51. Highest boiling point :-

(1) CH_3OH (2) CH_3Br
 (3) CH_3Cl (4) CH_4

52. Which Ion in excess NaOH does not form soluble complex :

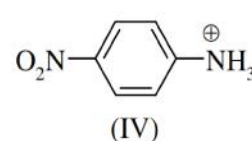
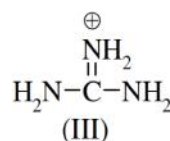
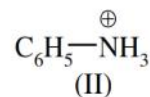
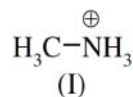
(1) Cr^{+3} (2) Zn^{+2} (3) Al^{+3} (4) Fe^{+3}

53. IUPAC name of the following compound is

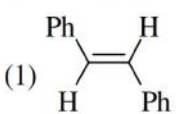
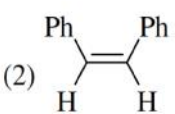


(1) Hex-2-en-5-yne (2) Hex-5-yn-2-ene
 (3) Hex-4-en-1-yne (4) Hex-1-yn-4-ene

54. Arrange the following in order of their decreasing acidic strength?



(1) $\text{I} > \text{II} > \text{III} > \text{IV}$ (2) $\text{IV} > \text{II} > \text{I} > \text{III}$
 (3) $\text{IV} > \text{III} > \text{II} > \text{I}$ (4) $\text{IV} > \text{I} > \text{II} > \text{III}$

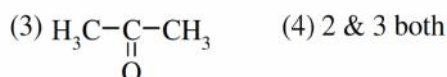
55. The conductivity of saturated solution of BaSO_4 is $3.06 \times 10^{-6} \text{ mho cm}^{-1}$ and its equivalent conductance is $1.53 \text{ mho cm}^2 \text{ eq}^{-1}$. Then K_{sp} for BaSO_4 will be
 (1) $4 \times 10^{-12} \text{ M}$ (2) $1 \times 10^6 \text{ M}$
 (3) $4 \times 10^{-6} \text{ M}^2$ (4) $1 \times 10^{-6} \text{ M}^2$
56. Xe has max. EN in :-
 (1) XeF_2 (2) XeF_4
 (3) XeO_3 (4) XeOF_2
57. Which is not a pseudohalide ?
 (1) $\text{CNO}^\ominus$ (2) $\text{RCOO}^\ominus$
 (3) $\text{OCN}^\ominus$ (4) $\text{NNN}^\ominus$
58. Which of the following is most reactive towards nucleophilic attack at carbonyl group
 (1) CH_3COCl (2) $\text{CH}_3\text{COOCH}_3$
 (3) CH_3CONH_2 (4) $\text{CH}_3\text{COOCOCH}_3$
59. No. of structural isomeric alkenes (molecular formula $= \text{C}_6\text{H}_{12}$) which all give n-hexane on hydrogenation in presence of metal catalyst
 (1) 2 (2) 3 (3) 4 (4) 5
60. Solid KClO_3 is taken in a container maintained at constant pressure of 1 atm. Upon heating following equilibrium is obtained
 $2\text{KClO}_3(\text{s}) \rightleftharpoons 2\text{KCl}(\text{s}) + 3\text{O}_2(\text{g})$
 If $\Delta H^\circ = 25 \text{ kcal/mol}$ and $\Delta S^\circ = 50 \text{ cal/K}$, at what temperature equilibrium will be established in the container. (Ignore variation of ΔH° and ΔS° with temperature.)
 (1) 298 K (2) 500 K
 (3) 5000 K (4) 300 K
61. Which of the following pair will have different oxidation number of oxygen ?
 (1) $\text{BaO}_2, \text{H}_2\text{O}_2$ (2) CO_2, SO_3
 (3) $\text{H}_2\text{O}_2, \text{O}_2\text{F}_2$ (4) $\text{KO}_2, \text{RbO}_2$
62. The number of hydrogen atoms directly attached to phosphorus atom in phosphorous acid is :-
 (1) 1 (2) 2
 (3) 3 (4) None
63. Which of the following hormones contains iodine
 (1) Thyroxine (2) Insulin
 (3) Testosterone (4) Adrenaline
64. In a crystalline solid, anion B are arranged in ccp lattice and cation A occupy 50% of the octahedral voids and 50% of tetrahedral voids what will be formula of crystalline solid?
 (1) AB (2) A_3B_2
 (3) A_2B_2 (4) A_2B_3
65. If bond enthalpies of $\text{N} \equiv \text{N}$, $\text{H}-\text{H}$ and $\text{N}-\text{H}$ bonds are x_1 , x_2 and x_3 respectively, ΔH_f° for NH_3 will be
 (1) $x_1 + 3x_2 - 6x_3$ (2) $\frac{1}{2}x_1 + \frac{3}{2}x_2 - 3x_3$
 (3) $3x_3 - \frac{1}{2}x_1 - \frac{3}{2}x_2$ (4) $6x_3 - x_1 - 3x_2$
66. How many types of F-S-F bonds are present in SF_4 :-
 (1) 2 (2) 4 (3) 3 (4) 5
67. During the extraction of iron, slag produced is :-
 (1) CO (2) FeSiO_3
 (3) MgSiO_3 (4) CaSiO_3
68. In Duma's method of estimation of nitrogen 0.35 g of an organic compound gave 55 ml of nitrogen collected at 300 K temp. and 715 mm pressure. The percentage of nitrogen in compound is (aqueous tension at 300 K = 15 mm)
 (1) 15.45 (2) 16.45 (3) 17.45 (4) 14.45
69. Which of the following decreases on dilution of electrolytic solution ?
 (1) Equivalent conductance
 (2) Molar conductance
 (3) Specific conductance
 (4) Conductance
70. Which will have minimum Osmotic pressure?
 (1) 200 mL 2M NaCl solution
 (2) 200 mL 1M glucose solution
 (3) 200 mL 2M urea solution
 (4) all have same
71. Which of the following is not hydrolysed ?
 (1) AsCl_3 (2) PF_3
 (3) SbCl_3 (4) NF_3
72. In the aluminothermite process aluminium acts as :-
 (1) An oxidising agent (2) Flux
 (3) A reducing agent (4) None of these
73. $\text{Ph}-\text{CH}=\text{CH}-\text{Ph} \xrightarrow[\text{CCl}_4]{\text{Cl}_2} \text{X} \xrightarrow{2\text{NaNH}_2} \text{Y}$
 $\xrightarrow[\text{BaSO}_4]{\text{H}_2+\text{Pd}} \text{Z}$
 Product Z is
 (1)  (2) 
 (3) $\text{Ph}-\text{C}(\text{Ph})=\text{CH}_2$ (4) $\text{Ph}-\text{C}\equiv\text{C}-\text{Ph}$

74. Henry's law constants for O_2 and N_2 are
 $K_{O_2} = 3.3 \times 10^7 \text{ atm}$, $K_{N_2} = 6.51 \times 10^7 \text{ atm}$

Calculate the ratio of $\frac{X_{O_2}}{X_{N_2}}$ i.e. the ratio of mole

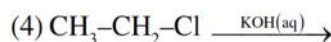
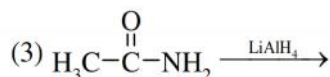
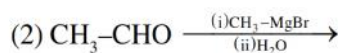
fractions of O_2 and N_2 dissolved in water at 25°C from air :-

- (1) 0.62 (2) 0.92
 (3) 0.31 (4) 0.53
75. The I.P. of a hydrogen like species is 36 eV. What is the value of second excitation energy
 (1) 5 eV (2) 12 eV
 (3) 32 eV (4) 27 eV
76. Which of the following molecules/ions are all the bonds not equal :-
 (1) ClF_3 (2) SiF_4 (3) XeF_4 (4) BF_4^-
77. Propanal on treatment with dilute sodium hydroxide forms
 (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO}$
 (2) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CHO}$
 (3) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}(\text{CH}_3)\text{CHO}$
 (4) $\text{CH}_3\text{CH}_2\text{COONa}$
78. Which of the following compound does not give positive Fehling's test



79. Number of moles of KMnO_4 reduced by 1 mol I^- in basic medium is :-
 (1) 3 (2) 2 (3) $2/3$ (4) $3/2$
80. pH of a solution obtained by mixing equal volume of 0.2 M NaOH & 0.2 M CH_3COOH ($K_a = 10^{-5}$) is :
 (1) 7 (2) 5 (3) 9 (4) 9.5
81. Which one of the following oxides is expected to exhibit paramagnetic behaviour ?
 (1) CO_2 (2) ClO_2
 (3) SO_2 (4) SiO_2
82. Which artificial sweetner contains chlorine?
 (1) Aspartame (2) Saccharin
 (3) Sucralose (4) Alitame

83. Alcohol is not formed as product in



84. For the reaction takes place at certain temperature
 $\text{NH}_2\text{COONH}_4(\text{s}) \rightleftharpoons 2\text{NH}_3(\text{g}) + \text{CO}_2(\text{g})$. If equilibrium pressure is 3X bar then $\Delta_r G^\circ$ would be

- (1) $-\text{RT} \ln 9 - 3\text{RT} \ln X$
 (2) $\text{RT} \ln 4 - 3\text{RT} \ln X$
 (3) $-3\text{RT} \ln X$
 (4) $-\text{RT} \ln 4 - 3\text{RT} \ln X$

85. Calculate volume of H_2 gas kept at STP if it contains as many H atoms as in 98 gm H_3PO_4 . [Atomic mass of P = 31]

- (1) 68.1 litre (2) 33.6 litre
 (3) 89.6 litre (4) 44.8 litre

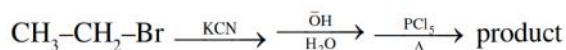
86. Which of the following not a square planar complex:

- (1) $[\text{AgF}_4]^-$ (2) $[\text{AuCl}_4]^-$
 (3) $[\text{RhCl}(\text{PPh}_3)_3]$ (4) $[\text{NiCl}_2(\text{PMe}_3)_2]$

87. Ozonolysis of an organic compound 'A' produces acetone and propionaldehyde in equimolar mixture. Identify A from the following compounds

- (1) 2-methyl-1-pentene (2) 1-pentene
 (3) 2-pentene (4) 2-Methyl-2-pentene

88. Major product of the following reaction



- (1) $\text{CH}_3-\text{CH}_2-\text{Cl}$ (2) $\text{H}_3\text{C}-\text{CH}_2-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{Cl}$
 (3) $\text{CH}_3-\text{CH}_2-\text{CN}$ (4) $\text{CH}_3-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{NH}_2$

89. Which of the following has minimum flocculation value for positively charged sol?

- (1) Cl^- (2) SO_4^{2-}
 (3) PO_4^{3-} (4) $[\text{Fe}(\text{CN})_6]^{4-}$

90. A x : 1 molar mixture of He and CH_4 is contained in a vessel at 20 bar pressure. Due to a hole in the vessel, the gas mixture leaks out. If the composition of the mixture effusing out initially is 8 : 1, then the value of x will be

- (1) 1 (2) 4 (3) 8 (4) 9

91. The plant group that possess spores and embryo but lack of vascular tissue and seeds are :-
 (1) Pteridophyta (2) Rhodophyta
 (3) Bryophyta (4) Phaeophyta
92. Given below are four method (A-D) and their modes of action (a-d) in achieving contraception. Select their correct matching.

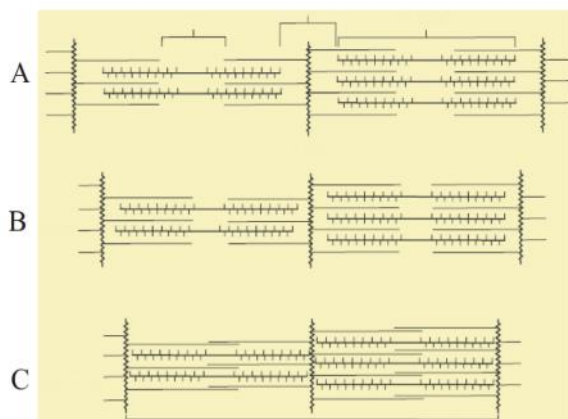
	Method		Mode of action
(A)	Mala-N	(a)	Prevent sperms reaching cervix
(B)	Today	(b)	Prevents ovulation
(C)	Cu7	(c)	Spermicides
(D)	Nirodh	(d)	Prevents Fertilization

- (1) (A)-(c), (B)-(d), (C)-(a), (D)-(b)
 (2) (A)-(b), (B)-(c), (C)-(a), (D)-(d)
 (3) (A)-(c), (B)-(a), (C)-(d), (D)-(b)
 (4) (A)-(b), (B)-(c), (C)-(d), (D)-(a)
93. If round yellow seed containing plants are cross pollinated then find out the no. of plants which is complete heterozygous in F_2 generation.
 (1) 9 (2) 3 (3) 3 (4) 4
94. Choose the incorrect match :-
 (1) Yeast – Bread & Beer production
 (2) *Agaricus* – Edible mushroom
 (3) *Amanita muscaria* – Hallucinogenic fungus
 (4) *Puccinia* – Smut fungus
95. Gibberellin was first extracted from :-
 (1) *Gibberella fujikuroi*
 (2) Algae
 (3) Bacteria
 (4) Roots of higher plants
96. Which glycosidic linkage is found at branching point in glycogen?
 (1) β 1 $\rightarrow$ 4
 (2) α 1 $\rightarrow$ 4
 (3) α 1 $\rightarrow$ 6
 (4) Both α 1 $\rightarrow$ 4 and α 1 $\rightarrow$ 6

97. DNA replication occurs during :
 (1) G_1 -Phase (2) G_2 -Phase
 (3) S-Phase (4) M-Phase
98. Which is oxidation state of iron in haemoglobin ?
 (1) Fe^- (2) Fe^{2+} (3) Fe^{3+} (4) Fe^{4+}
99. When an important predator starfish *Pisaster* was experimentally removed from an enclosed intertidal area of communities of the American pacific coast, it resulted in extinction of more than 10 species of invertebrates within a year. It demonstrates role of a predator as a factor for :-
 (1) preventing mutualism among prey species
 (2) reducing the intensity of competition among competing prey species
 (3) Over-exploitation of prey species
 (4) reducing reproductive success in prey species
100. If we completely remove decomposers from an ecosystem, the ecosystem functioning will be adversely affected because -
 (1) Mineral movement will be blocked
 (2) Herbivores will not receive solar energy
 (3) Energy flow will be blocked
 (4) Rate of decomposition of other components will be very high
101. In Gymnosperm pollination is mainly by -
 (1) Animals (2) Wind
 (3) Insect (4) Water
102. Which is **not correct** about amnion?
 (1) It provides a fluid medium to the developing embryo
 (2) It is formed by amniogenic cells and extra embryonic mesoderm
 (3) It takes part in the formation of human placenta
 (4) It is present in the amniota group of animals
103. *Cry* IAC and *cry* IIAb control
 (1) corn borer
 (2) cotton bollworm
 (3) *agrobacterium tumefaciens*
 (4) *E. coli*

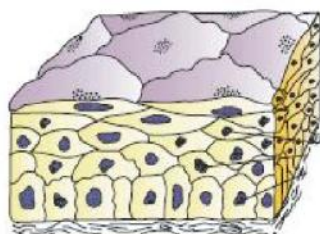
- 104.** Which of the following is **not** found in prokaryotic cells ?
 (1) Plasma membrane (2) Ribosomes
 (3) Mesosome (4) Nuclear envelope
- 105.** The oxygen produced during photosynthesis comes from water was proved by ?
 (1) Ruben, Hassid and Kamen
 (2) Robert Mayer
 (3) M. Calvin
 (4) Blackman
- 106.** Animal starch is :-
 (A) Glycogen
 (B) A branched polysaccharide
 (C) A polymer of α -glucose
 (D) Storage form of carbohydrate in animals
Options :
 (1) A and D (2) A and C
 (3) A, C and D (4) A, B, C and D
- 107.** Glyco proteins synthesized in :
 (1) Ribosome (2) Golgi apparatus
 (3) Plastid (4) Vacuole
- 108.** The factor which does not affect rate of alveolar diffusion ?
 (1) Solubility of gases
 (2) Thickness of membrane
 (3) Pressure gradient
 (4) Reactivity of gases
- 109.** How many statements are **correct**.
 (a) Net primary productivity is available biomass for consumption to heterotrophs
 (b) Secondary productivity is defined as rate of formation of new organic matter by heterotrophs
 (c) The annual net primary productivity of ocean is greater than land
 (d) Warm and moist environment favours decomposition
 (1) abcd (2) abc (3) abd (4) ac
- 110.** Less seasonal variation, undisturbed environment, long evolutionary time promotes :-
 (1) High species diversity in an area
 (2) High niche density in an area
 (3) Complex and stable ecosystem
 (4) All of the above
- 111.** In angiosperm free nuclear division takes place during ?
 (1) Gamete formation (2) Flower formation
 (3) Endosperm formation (4) Embryo formation
- 112.** Which of following is not the example of white fat ?
 (1) Hypodermis of skin
 (2) Hump of camel
 (3) Yellow bone marrow
 (4) In New born baby
- 113.** Gobargas is produces in the presence of which bacterium.
 (1) Methanogen
 (2) Staphylococcus aureas
 (3) E.coli
 (4) Pseudomonas
- 114.** Vascular cryptogames do not possess :-
 (1) Embryo (2) Tissue
 (3) Roots, stem & leaves (4) Seed
- 115.** The two pigment system theory of photosynthesis was proposed by ?
 (1) Hill (2) Blackman
 (3) Emerson (4) Arnon
- 116.** DNA replication is :-
 (1) Conservative and discontinuous
 (2) Conservative and semi-discontinuous
 (3) Semi-conservative and semi-discontinuous
 (4) Semi-conservative and continuous
- 117.** Which pair is wrong ?
 (1) Cisternae : E.R.
 (2) Oxisome : Mitochondria
 (3) Thylakoid : Leucoplast
 (4) Histones : Chromosome
- 118.** Prothrombin which helps in clotting of blood is released by ?
 (1) Monocyte (2) Erythrocyte
 (3) Lymphocyte (4) Blood platelets
- 119.** How many statements are **correct**.
 (a) In an aquatic ecosystem, grazing food chain is major conduits for energy flow
 (b) Detritus food chain may be connected with grazing food chain at some level
 (c) Each tropic level has a certain mass of living materials at a particular time called as standing crop
 (d) Species composition and stratification are two main structural features of an ecosystem
 (1) Two (2) Three (3) One (4) All

120. The figure given here represents three different conditions of sarcomeres. Identify these conditions and select the correct option :-



	A	B	C
(1)	Contraction	Relaxed	Maximum contracted
(2)	Relaxed	Contracted	Maximum contracted
(3)	Maximum contracted	Contracted	Relaxed
(4)	Relaxed	Contracted	Minimum contracted

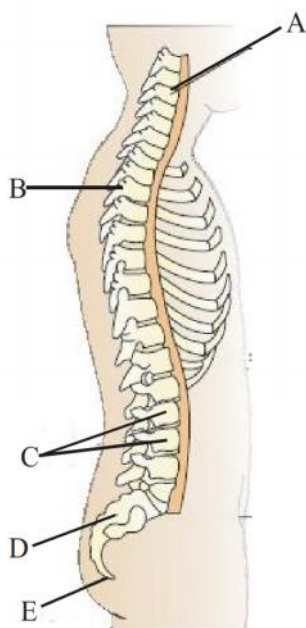
121. Secondary growth is the production of :-
 (1) New tissues from intercalary meristem
 (2) New conduction cells
 (3) New tissues from lateral meristem
 (4) New ground cells
122. Following diagram represent which type of epithelium ?



- (1) Simple squamous epithelium
 (2) Simple cuboidal epithelium
 (3) Simple columnar epithelium
 (4) Compound epithelium

123. Transgenic animal are use for :-
 (1) Study of disease
 (2) Biological product, vaccine safety
 (3) Both 1 and 2
 (4) None of these
124. Which one of the following statements about all the four of spongilla, Leech, Dolphin and Penguin is correct ?
 (1) All are bilaterally symmetrical
 (2) Penguin is homoiothermic while the remaining three are Poikilothermic
 (3) Leech is a fresh water form while all others are marine
 (4) Spongilla has special collard cells called choanocytes, not found in the remaining three.
125. Which of the following is associated with photo oxidation of water in photosynthesis ?
 (1) Cytochrome (2) Pigment system I
 (3) Pigment system II (4) Quinone
126. Haemoglobin molecule exhibits which level of protein structure?
 (1) Primary structure
 (2) Secondary structure
 (3) Tertiary structure
 (4) Quarternary structure
127. The cell wall of prokaryotes is made up of :
 (1) Cellulose (2) Glycolipid
 (3) Peptidoglycan (4) Lignin
128. pH of blood :-
 (1) is greater than 7
 (2) ranges between 7-8
 (3) is less than 7
 (4) None of these
129. Which of the following is a famous cause of biodiversity loss ?
 (1) Predation
 (2) Commensalism
 (3) Facultative association of species
 (4) Obligate interaction of species

130. Identify the part labelled as A to E in the given figure of a vertebral column and select the correct option :-



	A	B	C	D	E
(1)	Thoracic vertebra	Cervical vertebra	Lumbar vertebra	Sacrum	Coccyx
(2)	Thoracic	Cervical	Lumbar	Coccyx	Sacrum
(3)	Lumbar	Thoracic	Cervical	Coccyx	Sacrum
(4)	Cervical	Thoracic	Lumbar	Sacrum	Coccyx

131. Phloem parenchyma is absent in :-
 (1) Dicot stem
 (2) Dicot leaf
 (3) Monocot stem
 (4) Dicot root
132. Read the following statements carefully, which is incorrect about cockroach ?
 (1) In cockroach one pair testis present from 4 to 6th abdominal segments.
 (2) Mushroom shaped gland is present in the 6th - 7th abdominal segments
 (3) Sperms are stored in the seminal vesicles
 (4) The external genitalia are represented by phallic gland.

133. When a taxonomist uses the chemical constituents of the plant to resolve confusions, the he/she is dealing with :-

- (1) Cytotaxonomy
- (2) Numerical taxonomy
- (3) Chemotaxonomy
- (4) Alpha taxonomy

134. Which one of the following statement about certain given animals is correct?

- (1) Flatworms (Platyhelminthes) are coelomates
- (2) Round worms (Aschelminthes) are Pseudocoelomates
- (3) Molluscs are Acoelomates
- (4) Insects are Pseudocoelomates

135. Chlorophyll is soluble in :-

- (1) Water
- (2) Organic solvents
- (3) Water and organic solvents
- (4) Not in any solvent

136. 'Pomato' has been produced as the result of :-

- (1) Interspecific hybridisation between potato and tomato.
- (2) Somatic hybridisation
- (3) Mutation breeding
- (4) Recombinant DNA technique

137. The number of base pair of DNA in one nucleosome :

- (1) 146
- (2) 200
- (3) 54
- (4) 1200

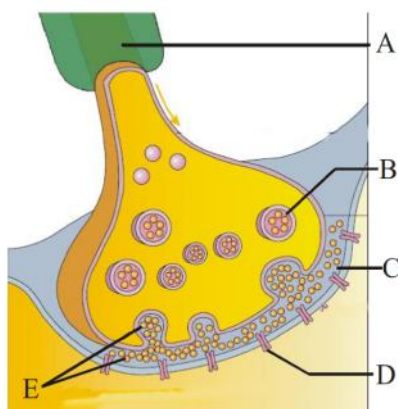
138. Which of the following age group is adolescent period :

- (1) 14-22 years
- (2) 12-15 years
- (3) 12-18 years
- (4) 14-24 years

139. Which of the following is **not** the case of parasitism in true sense ?

- (1) Hyperparasitism
- (2) Ectoparasitism
- (3) Endoparasitism
- (4) Brood parasitism

140. The given diagram shows axon terminal and synapse. Here A, B, C, D and E respectively represents ?



- (1) Axon terminal, synaptic cleft, synaptic vesicles, Neurotransmitters and receptors.
(2) Axon terminal, synaptic vesicles, synaptic cleft, receptors, and Neurotransmitters.
(3) Synaptic cleft, Synaptic vesicles, Axon terminals, Neurotransmitters, and receptors
(4) Synaptic cleft, axon terminal, synaptic vesicles, Neurotransmitters and receptors
141. Formation of which tissue is example of de-differentiation.
(1) Intercalary meristem
(2) Interfascicular cambium
(3) Intrafascicular cambium
(4) Apical meristem
142. Polycistronic mRNA is formed in Lac operon which is used in synthesis of :-
(1) β galactosidase permease
(2) β galactosidase transacetylase
(3) Both 1 and 2
(4) Only transacetylase
143. What is the number of broad or obligate taxonomic categories ?
(1) Five (2) Six (3) Seven (4) Two
144. Which of the following pairs are correctly matched.

Animal	Morphological features
(A) Crocodile	4 - Chambered heart
(B) Sea Urchin	Parapodia
(C) Obelia	Metagenesis
(D) Lemur	Thecodont
(1) Only A and B	(2) A, C and D
(3) B, C and D	(4) Only A and D

145. In kreb's cycle FAD participates as electron acceptor during conversion of :-
(1) Succinyl CoA to succinic acid
(2) Alfa keto glutarate to succinyl CoA
(3) Succinic acid to fumaric acid
(4) Fumaric acid to malic acid
146. Himgiri is a variety of wheat. It is resistant for :-
(1) White rust (2) Leaf rust
(3) Black rot (4) Leaf curl
147. Absorption of water, simple sugar and alcohol etc. takes place in :-
(1) Small intestine (2) Large intestine
(3) Stomach (4) Mouth
148. If regular dose of the drug is abruptly discontinued than tendency of body to manifest a characteristic and unpleasant withdrawal syndrome is called :
(1) Addiction (2) Dependence
(3) Allergy (4) Autoimmune disorder
149. Which one is almost **not** present in nature:-
(1) food chain (2) Food web
(3) both (4) None
150. Which of the following statement is correct regarding cerebellum ?
(1) Part of Hind brain
(2) It consist two cerebellar hemisphere and a vermis
(3) Arbor vitae is made up of white matter
(4) All of the above
151. In unilocular ovary with a single ovule the placentation is :-
(1) Axile (2) Marginal
(3) Basal (4) Free central
152. Hardy weinberg law is applicable when :-
(1) Mutation is absent
(2) Selection and migration is absent
(3) Both 1 and 2
(4) Only random drift
153. In a container bacteria are dividing by binary fission with the rate of 20 minutes. If initial number of bacteria in the container is 1.75×10^{32} , then how many bacteria are produced in 2 hours ?
(1) 112×10^{30} (2) 64×10^{32}
(3) 112×10^{32} (4) $.64 \times 10^{30}$

154. What is common between Parrot, Platypus and Kangaroo?
- Ovoparity
 - Homoiothermy
 - Tooth less jaws
 - Functional Post - anal tail
155. If the frequency of an allele reaches 1.0, the allele is said to be:-
- lost
 - stable
 - perfect
 - fixed
156. Biofortification refers to breeding crops for :-
- Disease resistance
 - Pest resistance
 - Improved nutritional quality
 - Faster yield
157. Read the following statements carefully :-
- Gastro-oesophageal sphincture regulates the opening of oesophagus into the stomach
 - The stomach is located in the lower left portion of abdominal cavity
 - The opening of the stomach into duodenum is guarded by the cardiac sphincture
 - Small intestine has 3 region :- 'U'-shaped jejunum, a long coiled middle portion duodenum and a highly coiled ileum.
- On the basis of above statements which statements are not true ?
- Statments 'a', 'c' and 'd'
 - Statments 'b', 'c' and 'd'
 - Statments 'a', 'b' and 'c'
 - Only statement 'b' and 'd'
158. Opioids are the drugs which bind to specific ____ (A) ____ receptors present in ____ (B) ____.
- What is true about A and B
- A-CNS, B-Opioid
 - A-Cannabinoids, B-Central nervous system
 - A-Central nervous system, B-Cannabinoids
 - A-Opioid, B-Central nervous system

159. Producer → Consumer → Consumer → Consumer
- (1) (2) (3)

In a aquatic food chain a disease reduced the no. of consumer 2. What effect would be likely to have on the no. of consumer (1) and (3).

	Consumer (1)	Consumer (3)
(1)	↑ in No.	↑ in No.
(2)	↑ in No.	↓ in No.
(3)	↓ in No.	↑ in No.
(4)	↓ in No.	↓ in No.

160. The structure in a human body that assist in body balance are located in :-
- Outer ear
 - Inner ear
 - Middle ear
 - Eustachian tube
161. What is the eye of potato ?
- Axillary bud
 - Accessory bud
 - Adventitious bud
 - Apical bud
162. Homozygous golden leaves containing plants of snapdragon are never obtained by which of the following cross.
- GG × gg
 - gg × Gg
 - Gg × Gg
 - gg × gg
163. Auxospores (rejuvenating spores) are formed in :-
- Dinoflagellates
 - Diatoms
 - Slime moulds
 - Euglenoids
164. A solution of 1.0 M glucose develops a pressure of 27 bars in an osmometer. What is **not correct** ?
- Pressure potential is -27 bars
 - Osmotic pressure is 27 bars
 - Osmotic potential is -27 bars
 - Solute potential is -27 bars
165. Match the following and select correct answer :-

(A)	Coal deposits	(a)	First mammal
(B)	Ichthyosaurs	(b)	Daggers like teeth
(C)	Tyrannosaurs	(c)	Fish like reptile
(D)	Shrew	(d)	Giant ferns

- A-(d), B-(c), C-(b), D-(a)
- A-(d), B-(c), C-(a), D-(b)
- A-(c), B-(d), C-(b), D-(a)
- A-(a), B-(b), C-(c), D-(d)

166. Which of the following combinations is correct?
- (1) Amylase – Peptide bond
 - (2) Protease – Phosphodiester bond
 - (3) Lipase – Ester bond
 - (4) Nucleases – Glycosidic bond
167. Which of the following statement is **correct** ?
- (1) The contraction of internal intercostal muscles lifts up the ribs and sternum
 - (2) The RBCs transport oxygen only
 - (3) The thoracic cavity is anatomically an air tight chamber
 - (4) Healthy man can inspire approximately 500 ml of air per minute
168. Which of the following is not responsible for ringworm :
- (1) Trichophyton
 - (2) Microsporum
 - (3) Entamoeba histolytica
 - (4) Epidermophyton
169. Which is **true** for primary succession :-
- (a) Humus or organic matter is absent in early stage
 - (b) The area is barren from beginning
 - (c) Reproductivity structures of previous living beings may be present
 - (d) Primary succession is slower than secondary succession
- (1) ab
 - (2) abc
 - (3) abd
 - (4) cd
170. Which one of the following is correct difference between rod cells and cone cells of our retina ?

	Rod cells	Cone cells
(1)	Vision in poor light	Colour vision
(2)	Concentrated in centre of Retina	Distributed all over retina
(3)	Low in number	High in number
(4)	Iodopsin	Rhodopsin

171. The blastomeres in the blastocyst are arranged into an outer layer called ___A___, and inner group of cells attached to trophoblast called the ___(B)___.
- (1) 'A' is trophoblast and 'B' are inner cells mass
 - (2) Both are inner cells mass
 - (3) Both are trophoblast
 - (4) 'A' are inner cells mass and 'B' is trophoblast
172. Parental genotype ratio in F_2 generation by crossing pea plant of Genotype $RRYY \times rryy$
- (1) 2 : 2 : 4
 - (2) 1 : 1
 - (3) 1 : 1 : 1 : 1
 - (4) 1 : 2 : 2 : 4 : 1 : 2 : 1 : 2 : 1
173. A prokaryotic autotrophic nitrogen fixing symbiont is found in :-
- (1) *Alnus*
 - (2) *Pisum*
 - (3) *Cicer*
 - (4) *Cycas*
174. Apical dominance is caused by the production of :-
- (1) Gibberellin in lateral buds
 - (2) Cytokinins in stem tips
 - (3) Auxins in stem tip
 - (4) Absciscic acid in lateral buds
175. Choose the incorrect match regarding man and their characters :-

(1)	Dryopithecus and Ramapithecus	like gorilla
(2)	Australopithecus	Lived in East African grassland
(3)	Homo habilis	First human like being the hominid
(4)	Homo erectus	Arose during ice age between 75,000-10,000 years age

176. The semi-conservative replication of DNA was proved by using which isotopes?
- (1) N^{14} and N^{15}
 - (2) N^{14} and P^{31}
 - (3) P^{31} and C^{14}
 - (4) N^{14} and C^{14}
177. In human which of the following is **not** a step in respiration ?
- (1) Alveolar diffusion of O_2 and CO_2
 - (2) Transport of gases by blood
 - (3) Diffusion of O_2 and CO_2 between blood and tissue
 - (4) Utilization of CO_2 by cells for catabolic reaction

178. Gestation period in elephant is :-
 (1) 32 week (2) 21-22 month
 (3) 21-22 week (4) 16-18 month
179. How many statements are **correct** :-
 (a) Pyramid of biomass in sea is also generally inverted
 (b) pyramid of energy is never inverted
 (c) In terrestrial ecosystem a much larger fraction of energy flows through detritus food chain
 (d) Humus is highly resistant to microbial action and undergoes decomposition at an extremely slow rate
 (1) One (2) Two
 (3) Three (4) All of these
180. Given below is an incomplete table about certain hormones, select the option that correctly fills the blanks A, B and C :-

Glands	Secretion	Effect on body
A	Oestrogen	Maintain secondary sexual characters
α -Cells of islets of Langerhans	B	Raises blood sugar level
Anterior pituitary	C	Oversecretion leads to gigantism

	A	B	C
(1)	Placenta	Glucagon	Calcitonin
(2)	Ovary	Glucagon	Growth hormone
(3)	Placenta	Insulin	Vasopressin
(4)	Ovary	Insulin	Calcitonin